Introduction to the course

CSE 451
Advanced Topics in Computer Science
Fall 2023
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CS 451

Topics in Computer Science

- Topic this term: Data Science

- This course will provide an introduction to the main topics in data science and knowledge discovery, including exploratory data analysis, data cleaning, model building, and model evaluation.

Instructor

Mariam El Mezouar, PhD

- Office: Girouard 340
- Email: mariam.el-mezouar@rmc.ca
- Website: https://mariam-elmezouar.github.io/

Meeting times

Day	Time	What	Where
Tuesday	08:00 - 09:50	Lab	Girouard 332
Wednesday	13:40 - 16:30	Lecture	Girouard 332

Course Material

- All course material will be made available on the instructor's website https://mariam-elmezouar.github.io/ under the tab CSE451.

- Direct link to course: https://mariam-elmezouar.github.io/CSE451/index.html
- Link also posted on Moodle.

Learning objectives

After taking this course, the students should be able to:

- **Develop Proficiency in Data Analysis**: apply proper techniques for the exploration, cleaning, and preparation of data.
- **Apply Advanced Machine Learning**: apply advanced machine learning algorithms, models, and evaluation methods to solve real-world data-driven problems effectively.
- **Leverage Data for Informed Decision-Making**: extract actionable insights from data and make data-informed decisions that drive value in various domains and industries.

Prerequisites

- The prerequisites of this course include CSE101 (Algorithms) and MAE209 (Prob and stats).

- Data challenges, labs, and practice exercises will require the use of Python/Jupyter notebook.

Grading Policy

Assessment Type	**Weight**	**Quantity**
Midterm exam	20%	1
Data Challenges	20%	4
Topic Presentation	10%	1
Project	50%	1

Topics

Week	Topic	Assessments
1	Introduction	
2	Data exploration	
3	Data preparation	Data challenge 1
4	Model building and evaluation - Classification	
5	Model building and evaluation - Regression	Data challenge 2
6	Model building and evaluation - Clustering	
7	----	Midterm
8	Beyond numbers: Natural Language Processing	
9	Beyond numbers: Computer vision	Data challenge 3
10	Students topic presentations	
11	Advanced topic 1	Data challenge 4
12	Advanced topic 2	
13	Project presentations	

Course Assessments: Data Challenges

- Will take place on the dates indicated on the website (the Wednesday sessions).
- Duration: will vary – up to 50 minutes.
- Format: Data analysis task on a Jupyter notebook.
- Individual.
- Access to courses notes and internet permitted (minus LLMs – gpt or others).
- No collaboration with other students.
- Submission: Submit Jupyter notebook to Moodle.

Course Assessments: Midterm

- The tentative date is October 18th (to be confirmed at the beginning of October)
- Duration: 110 minutes.
- On computer.

Course Assessments: Project

- A dataset will be provided.
- During the first lab sessions, you will be asked to perform a set of tasks on that data (e.g. data exploration and cleaning).
- After that, I will sit individually with each of you and discuss some questions that we can ask from that data, and the types of models that can be applied to get those questions answered.
- More details will be provided later.

Course Assessments: Presentation

- A data science related topic to be selected by you and approved by me.
- A 30 minute presentation on that topic + 20 minute discussion.

Next…

Introduction to Data Science